Perform the following operations using Python on any open source dataset (e.g., data.csv)

1. Import all the required Python Libraries.
2. Locate an open source data from the web (e.g., https://www.kaggle.com). Provide a clear description of the data and its source (i.e., URL of the web site).
3. Load the Dataset into pandas dataframe.
4. Data Preprocessing: check for missing values in the data using pandas isnull(), describe() function to get some initial statistics. Provide variable descriptions. Types of variables etc. Check the dimensions of the data frame.
5. Data Formatting and Data Normalization: Summarize the types of variables by checking the data types (i.e., character, numeric, integer, factor, and logical) of the variables in the data set. If variables are not in the correct data type, apply proper type conversions.
6. Turn categorical variables into quantitative variables in Python.

```
#Importing necessary libraries
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

#Reading the dataset
df = pd.read_csv("Auto.csv")
df.head()
```

```
    mpg  cylinders  displacement  horsepower  weight  acceleration
year  \
0  18.0          8         307.0       130.0  3504.0          12.0
70.0
1  15.0          8         350.0       165.0  3693.0          11.5
70.0
2  18.0          8         318.0       150.0  3436.0          11.0
70.0
3  16.0          8         304.0       150.0  3433.0          12.0
70.0
4  17.0          8         302.0       140.0  3449.0          10.5
70.0

   origin                       name
0     1.0  chevrolet chevelle malibu
1     1.0          buick skylark 320
2     1.0         plymouth satellite
3     1.0              amc rebel sst
4     NaN                ford torino
```

```
df.tail()
```

```
      mpg  cylinders  displacement  horsepower  weight  acceleration
year  \
387  27.0          4         140.0        86.0  2790.0          15.6
82.0
```

```
388  44.0          4          97.0        52.0  2130.0          24.6
82.0
389  32.0          4         135.0        84.0  2295.0          11.6
82.0
390  28.0          4         120.0        79.0  2625.0          18.6
82.0
391  31.0          4         119.0        82.0  2720.0          19.4
82.0

     origin             name
387     1.0  ford mustang gl
388     2.0        vw pickup
389     1.0    dodge rampage
390     1.0      ford ranger
391     1.0       chevy s-10
```

```
print("\nStatistical Summary:")
#Descriptive statistics
print(df.describe())
```

```
Statistical Summary:
              mpg   cylinders  displacement  horsepower
weight  \
count  392.000000  392.000000    392.000000  390.000000   391.000000

mean    23.445918    5.471939    194.411990  104.471795  2978.365729

std      7.805007    1.705783    104.644004   38.584571   850.349703

min      9.000000    3.000000     68.000000   46.000000  1613.000000

25%     17.000000    4.000000    105.000000   75.000000  2224.500000

50%     22.750000    4.000000    151.000000   92.500000  2807.000000

75%     29.000000    8.000000    275.750000  128.000000  3616.500000

max     46.600000    8.000000    455.000000  230.000000  5140.000000


       acceleration        year      origin
count    391.000000  391.000000  387.000000
mean      15.489003   75.994885    1.583979
std        2.853878    3.675975    0.808024
min        0.500000   70.000000    1.000000
25%       13.700000   73.000000    1.000000
50%       15.500000   76.000000    1.000000
75%       17.000000   79.000000    2.000000
max       24.800000   82.000000    3.000000
```

```
#Printing Shapes and Size of the dataset
print("\nShape of the dataset:", df.shape)
print("Size of the dataset:", df.size)
```

```
Shape of the dataset: (392, 9)
Size of the dataset: 3528
```

```
#Checking for missing values
print("\nMissing Values:")
print(df.isnull().sum())
```

```
Missing Values:
mpg              0
cylinders        0
displacement     0
horsepower       2
weight           1
acceleration     1
year             1
origin           5
name             0
dtype: int64
```

```
#Handling missing values by filling them with the mean of the
respective columns
df.fillna(df.mean(numeric_only=True), inplace=True)
print(df.isnull().sum())
```

```
mpg              0
cylinders        0
displacement     0
horsepower       0
weight           0
acceleration     0
year             0
origin           0
name             0
dtype: int64
```

```
df.head()
```

```
    mpg  cylinders  displacement  horsepower  weight  acceleration
year  \
0  18.0          8         307.0       130.0  3504.0          12.0
70.0
1  15.0          8         350.0       165.0  3693.0          11.5
70.0
2  18.0          8         318.0       150.0  3436.0          11.0
70.0
```

```
3  16.0          8         304.0       150.0  3433.0          12.0
70.0
4  17.0          8         302.0       140.0  3449.0          10.5
70.0

     origin                       name
0  1.000000  chevrolet chevelle malibu
1  1.000000          buick skylark 320
2  1.000000         plymouth satellite
3  1.000000              amc rebel sst
4  1.583979                ford torino
```

```
print("\nShape of Data:", df.shape)
print("Total Elements:", df.size)
```

```
Shape of Data: (392, 9)
Total Elements: 3528
```

```
#Displaying the data types of each column
print(df.dtypes)
```

```
mpg              float64
cylinders          int64
displacement     float64
horsepower       float64
weight           float64
acceleration     float64
year             float64
origin           float64
name              object
dtype: object
```

```
#Converting categorical variables to numerical variables
if 'engine-location' in df.columns:
    df['engine-location'] = df['engine-location'].astype('category')

if 'horsepower' in df.columns:
    df['horsepower'] = pd.to_numeric(df['horsepower'],
errors='coerce')

numeric_cols = df.select_dtypes(include=np.number).columns

#Normalizing the numeric columns using Min-Max scaling
for col in numeric_cols:
    df[col] = (df[col] - df[col].min()) / (df[col].max() -
df[col].min())

print("\nAfter Normalization:")
df.head()
```

```
After Normalization:

        mpg  cylinders  displacement  horsepower    weight
acceleration  \
0  0.239362        1.0      0.617571    0.456522  0.536150
0.473251
1  0.159574        1.0      0.728682    0.646739  0.589736
0.452675
2  0.239362        1.0      0.645995    0.565217  0.516870
0.432099
3  0.186170        1.0      0.609819    0.565217  0.516019
0.473251
4  0.212766        1.0      0.604651    0.510870  0.520556
0.411523

   year   origin                        name
0   0.0  0.00000  chevrolet chevelle malibu
1   0.0  0.00000          buick skylark 320
2   0.0  0.00000         plymouth satellite
3   0.0  0.00000              amc rebel sst
4   0.0  0.29199                ford torino
```

```
#Converting categorical variables to numerical variables
df_encoded = pd.get_dummies(df)
df_encoded.head()
```

```
        mpg  cylinders  displacement  horsepower    weight
acceleration  \
0  0.239362        1.0      0.617571    0.456522  0.536150
0.473251
1  0.159574        1.0      0.728682    0.646739  0.589736
0.452675
2  0.239362        1.0      0.645995    0.565217  0.516870
0.432099
3  0.186170        1.0      0.609819    0.565217  0.516019
0.473251
4  0.212766        1.0      0.604651    0.510870  0.520556
0.411523

   year   origin  name_amc ambassador brougham  name_amc ambassador
dpl  ...  \
0   0.0  0.00000                          False
False  ...
1   0.0  0.00000                          False
False  ...
2   0.0  0.00000                          False
False  ...
3   0.0  0.00000                          False
False  ...
```

```
4   0.0  0.29199                          False
False  ...

   name_volvo 145e (sw)  name_volvo 244dl  name_volvo 245  name_volvo
264gl  \
0                 False             False           False
False
1                 False             False           False
False
2                 False             False           False
False
3                 False             False           False
False
4                 False             False           False
False

   name_volvo diesel  name_vw dasher (diesel)  name_vw pickup  name_vw
rabbit  \
0              False                    False           False
False
1              False                    False           False
False
2              False                    False           False
False
3              False                    False           False
False
4              False                    False           False
False

   name_vw rabbit c (diesel)  name_vw rabbit custom
0                      False                  False
1                      False                  False
2                      False                  False
3                      False                  False
4                      False                  False

[5 rows x 309 columns]
```

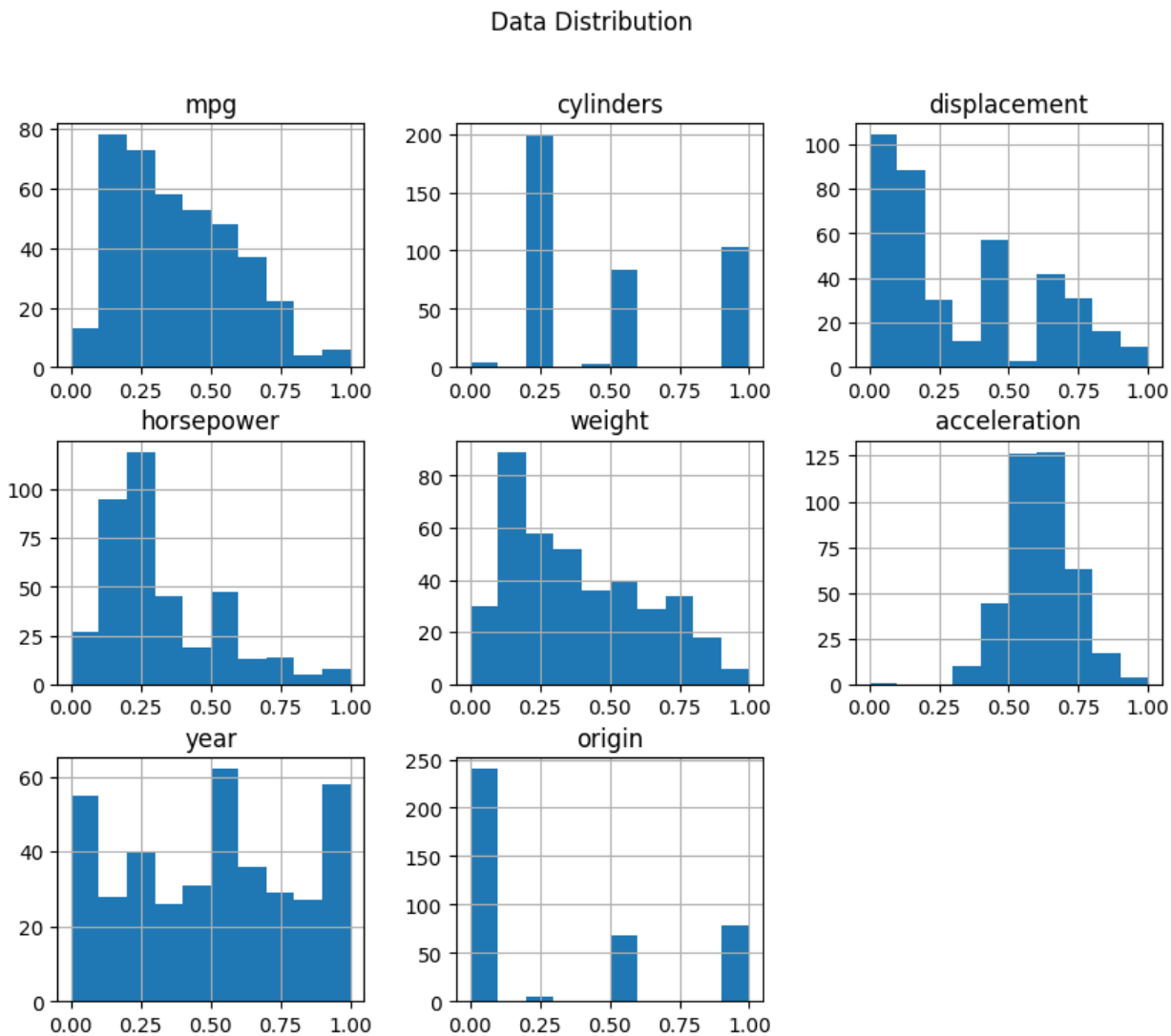
```python
print("\nAfter Encoding:")
print(df_encoded.head())
```

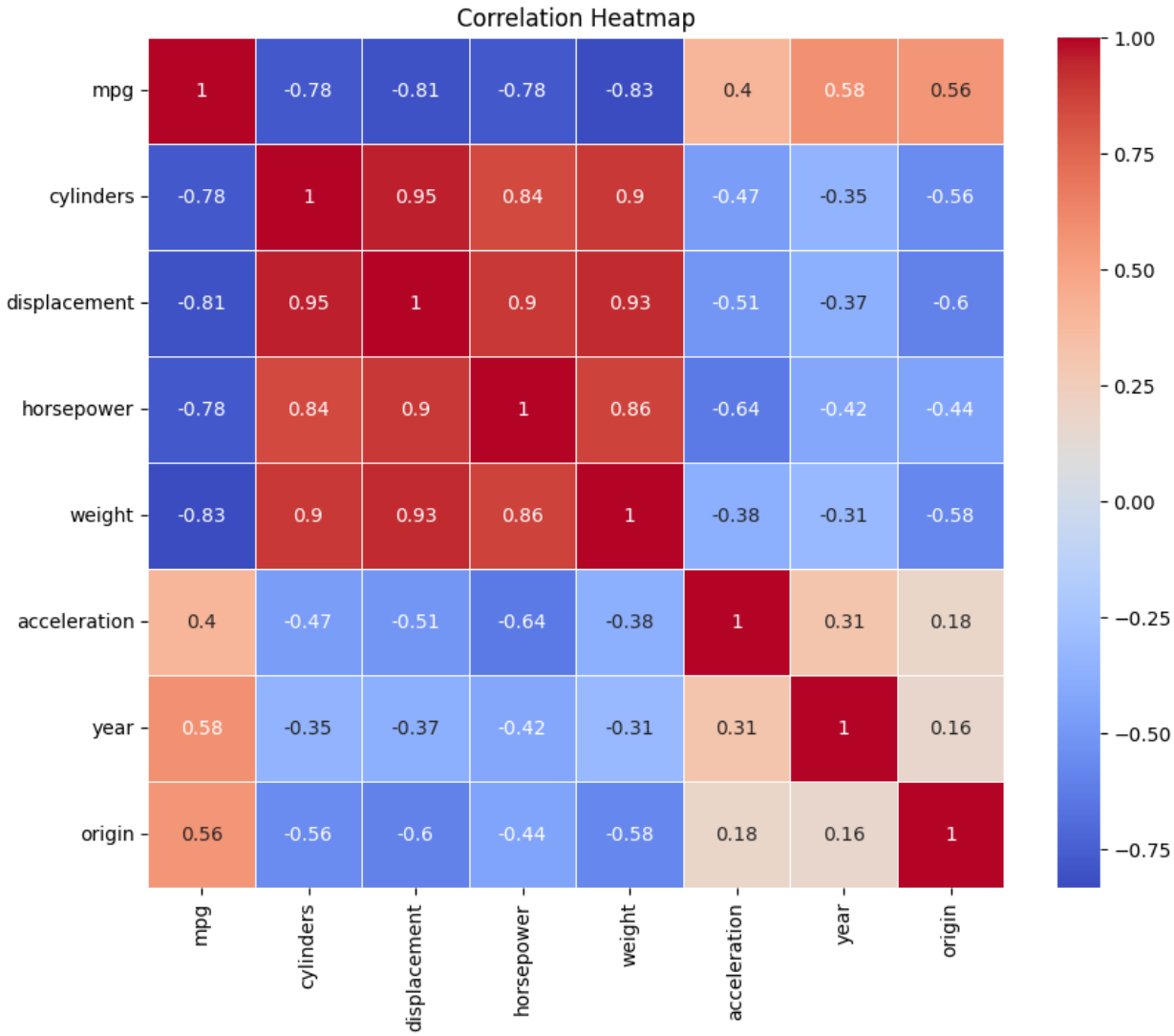
```
After Encoding:
        mpg  cylinders  displacement  horsepower    weight
acceleration  \
0  0.239362        1.0      0.617571    0.456522  0.536150
0.473251
1  0.159574        1.0      0.728682    0.646739  0.589736
0.452675
2  0.239362        1.0      0.645995    0.565217  0.516870
```

```
0.432099
3  0.186170        1.0      0.609819    0.565217  0.516019
0.473251
4  0.212766        1.0      0.604651    0.510870  0.520556
0.411523

   year   origin  name_amc ambassador brougham  name_amc ambassador
dpl  ...  \
0   0.0  0.00000                          False
False  ...
1   0.0  0.00000                          False
False  ...
2   0.0  0.00000                          False
False  ...
3   0.0  0.00000                          False
False  ...
4   0.0  0.29199                          False
False  ...

   name_volvo 145e (sw)  name_volvo 244dl  name_volvo 245  name_volvo
264gl  \
0                 False             False           False
False
1                 False             False           False
False
2                 False             False           False
False
3                 False             False           False
False
4                 False             False           False
False

   name_volvo diesel  name_vw dasher (diesel)  name_vw pickup  name_vw
rabbit  \
0              False                    False           False
False
1              False                    False           False
False
2              False                    False           False
False
3              False                    False           False
False
4              False                    False           False
False

   name_vw rabbit c (diesel)  name_vw rabbit custom
0                      False                  False
1                      False                  False
2                      False                  False
3                      False                  False
```

```
4                          False                    False

[5 rows x 309 columns]
```

```
#Visualizing the distribution of the data
df_encoded.hist(figsize=(10,8))
plt.suptitle("Data Distribution")
plt.show()
```




```
#plotting heatmap for numerical columns
plt.figure(figsize=(10,8))
new_df = df_encoded.select_dtypes(include=np.number)
sns.heatmap(new_df.corr(), annot=True, cmap='coolwarm',
linewidths=0.5)
plt.title("Correlation Heatmap")
```

```
Text(0.5, 1.0, 'Correlation Heatmap')
```